Exercises: Linear Algebra for ML

November 2, 2026

Exercises

Ex. 1) Given two vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ in $\mathbb{R}^n$.

a) Show that the vectors $\mathbf{u}_1$ and $\mathbf{u}_2$ defined by:

$$\mathbf{u}_1 = \mathbf{a}_1 \tag{1}$$

$$\mathbf{u}_2 = \mathbf{a}_2 - \frac{\mathbf{u}_1^\top \mathbf{a}_2}{\mathbf{u}_1^\top \mathbf{u}_1}\mathbf{u}_1 \tag{2}$$

are orthogonal to each other.

b) Show that if $\mathbf{a}_2$ is a multiple of $\mathbf{a}_1$, the previous procedure produces a zero vector for $\mathbf{u}_2$.

c) Show by induction that for any $k \leq n$ linearly independent vectors $\mathbf{a}_1, \ldots, \mathbf{a}_k$, the vectors $\mathbf{u}_i$, $i = 1, \ldots k$, are orthogonal, where

$$\mathbf{u}_i = \mathbf{a}_i - \sum_{j=1}^{i-1} \frac{\mathbf{u}_j^\top \mathbf{a}_i}{\mathbf{u}_j^\top \mathbf{u}_j}\mathbf{u}_j. \tag{3}$$

d) What do we call the procedure of constructing orthogonal vectors from nonorthogonal ones?

Ex. 2) The expression $\alpha \boldsymbol{u}$ for $\alpha \in \mathbb{R}$ and unit vector $\boldsymbol{u} \in \mathbb{R}^n$ defines a line of points that may be obtained by varying the value of α.

a) Derive an expression for the point $\boldsymbol{y}$ that lies on this line that is as close as possible to an arbitrary point $\boldsymbol{x} \in \mathbb{R}^n$.

b) What do we call the operation of replacing a point by its nearest member within some set?

Ex. 3) For a square matrix $\mathbf{A}$ of size $n \times n$, a vector $\mathbf{u}_i \neq 0$ which satisfies

$$\mathbf{A}\mathbf{u}_i = \lambda_i \mathbf{u}_i \tag{4}$$

is called a eigenvector of $\mathbf{A}$, and λ_i is the corresponding eigenvalue. For a matrix of size $n \times n$, there are n eigenvalues λ_i (which are not necessarily distinct).

(a) Show that if $\mathbf{u}_1$ and $\mathbf{u}_2$ are eigenvectors with $\lambda_1 = \lambda_2$, then $\mathbf{u} = \alpha\mathbf{u}_1 + \beta\mathbf{u}_2$ is also an eigenvector with the same eigenvalue.

(b) Assume that none of the eigenvalues of $\mathbf{A}$ is zero. Denote by $\boldsymbol{U}$ the matrix where the column vectors are linearly independent eigenvectors $\mathbf{u}_i$ of $\mathbf{A}$. Verify that (**??**) can be written in matrix form as $\mathbf{A}\boldsymbol{U} = \boldsymbol{U}\boldsymbol{\Lambda}$, where $\boldsymbol{\Lambda}$ is a diagonal matrix with the eigenvalues λ_i as diagonal elements.

Ex. 4) a) Show that[1]

$$\mathrm{tr}(\mathbf{A}) =: \sum_i A_{ii} = \sum_i \lambda_i.$$

b) Show that [2]

$$\det \mathbf{A} = \prod_i \lambda_i.$$

Ex. 5) **The pseudoinverse and a minimum-norm solution.** Let

$$A = \begin{pmatrix} 1 & 1 \end{pmatrix}, \qquad b = 2.$$

For a matrix with linearly independent rows, the Moore–Penrose pseudoinverse is $A^+ = A^\top (AA^\top)^{-1}$.

a) Compute A^+ and verify that $AA^+ = 1$.

b) Use $x_\star = A^+ b$ to find a solution of $Ax = b$.

c) Write every solution as $x = (t, 2-t)^\top$, where $t \in \mathbb{R}$. Show that $x_\star$ has the smallest Euclidean norm among all solutions.

[1] You can use $\mathrm{tr}(\mathbf{AB}) = \mathrm{tr}(\mathbf{BA})$.

[2] Use $\det \mathbf{A}^{-1} = 1/(\det \mathbf{A})$ and $\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B})$ for any $\mathbf{A}$ and $\mathbf{B}$.